Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/25/2005
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Solid Waste Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	6
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Alleman , Jim (Purdue Univesity) Glass, Charles (Howard University) Volenec, Jeffrey (Purdue University) Joern, Brad (Purdue University) Brown, Paul (Purdue University) Turco, Ron (Purdue University) Nies, Larry (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The air/water and solids focus groups of the NSCORT were merged into a single resource-recovery (R2) focus area during year 3 of Center operations, because many R2 projects are now coming together across disciplines. Two research projects came to a close during the past year, a new project started up, and a center-wide analytical effort was initiated within the R2 group. In addition, sensitivity has spread across the Center regarding the need for all research projects to identify steps or points in bioprocessing where hazards to human crews might occur. Potential hazards resulting from biological approaches to bioprocessing include microbial contamination, toxic metabolites in waste feedstocks and/or bioreactor products, and noxious gases released from metabolic or physico-chemical degradation of organic substances in air, water, or solid phases. Absorption, inhalation, or ingestion of hazardous organisms or substances by crews are of particular concern in closed systems.
A new NSCORT approach to degrading crop and food waste (Plant-Based Anaerobic-Aerobic Bioreactor-Linked Operation, or PAABLO) biologically pretreats grey water, reduces solid waste loads, and produces methane for subsequent energy reclamation, without producing copious waste heat. An analogous bio-reactor (Waste-AABLO, or WAABLO) reduces human solid waste in a separate process stream. PAABLO recalcitrant residues are directed to a fungal composter for partial hydrolysis of ligno-cellulose, the hydrolysates are returned to the anaerobic reactor to maximize methanogenesis, energy reclamation, and solids reduction, and the pre-treated grey water is directed to BREATHe I for aerobic co-treatment of air and water. BREATHe I also can remove NH4+ from aquarium water and urea from urine in separate process streams. Micro-, nano-, and RO-membranes filter out trace organics and desalt inorganics from BREATHe effluents, and iodine and UVc provide final polishing and fail-safe quality control. Noxious gases emitted from bioreactors and incinerators are processed by BREATHe II biofilters and/or concentrated onto zeolite adsorbent particles. Nitrogen and sulfur-rich adsorbents or eluates are added to fungal composters and microbial bioreactors to stimulate the growth of reactor organisms feeding on N- and S-deficient ligno-cellulose and enhance their hydrolysis of feedstock polymers. Consumption by Tilapia fish of microbial biomass containing amino acids essential for fish development enables those fish to indirectly reduce the load of recalcitrant crop waste while supplementing crew diets with small amounts of animal protein.

By using products of one waste treatment as substrates for another, by focusing on bioregenerative approaches to primary processing and limiting physico-chemical approaches to secondary processing and polishing functions, and by analyzing for bio-hazards at critical process points, the NSCORT R2 group is significantly reducing ESM for resource recovery while minimizing potential risks in closed systems.

Cary Mitchell for Al Heber, R2 Focus Area Lead STAR associated objectives are: The primary goal of this project is to develop a novel high-temperature solids digestion system for processing biodegradable wastes within a sustainable closed-loop ecosystem. The pathogen-free residuals streams generated by this process would be amenable to direct water and nutrient recovery. The design is loosely modeled after the successful, commercial Autothermal Thermophilic Aerobic Digestion (ATAD) wastewater sludge treatment process (1). Associated objectives are: 1-To create a thermophilic reactor system mechanically able to suitable mix and aerate high input solids levels at an expected range of 6-10% solids, 2-To maximize the positive ESM-enhancing attributes in terms of gas transfer rates to maximize performance, solids shear to reduce particle size, solids and organic destruction performance to reduce waste volume and enable further recovery of water, nutrients and carbon, and pathogen pasteurization to prepare the product for further use or storage (2,3), 3-To minimize ESM-degrading attributes, in regard to mass and volume, crewtime, noise generation, vibration, energy consumption, and heat loss, 4-To evaluate the system’s performance utilizing a long-term mission waste stream including human fecal, paper, food residuals, and plant materials, and 5-To evaluate the complementary use and operational performance of a vacuum waste collection system, particularly in regard to high-solids waste transfer.

For PAABLO/WAABLO:

This project will focus on adaptation and use of aerobic-anaerobic technologies for the treatment of three sources of waste (plant residues, greywater and human waste) on the Mars colony. We are proposing the development of dual systems, one to handle the combination of plant waste and greywater and one to handle the human waste. While both will result in the formation of CH4, the plant waste-grey water system (e.g., PAABLO) will also be used to develop an organic plant growth media similar to compost. Human waste treatment system (WAABLO) use the same approach but the resulting biosolids are not used in food production. In both cases, the anaerobically processed waste streams will provide a renewable energy source and will result in a reduction in the mass plant waste present. In the long-term the anaerobic digestor will be a source of methane (leading to small scale electricity generation), produce water, treat waste materials while conserving O2 and act as a heat sink as optimal digestor operation is achieved with an elevated temperature. Solid waste from the PAABLO system will be an ideal bedding material for the production of mushrooms and any other plant types that are difficult to grow in liquid systems. In order to maintain a high quality food, at no time do we propose using biosolids from WAABLO in food production. These anaerobic digestor systems will provide a robust treatment technology that will allow a rapid and consistent conversation of waste to energy and in the case of PAABLO needed secondary materials.

Nitrogen Cycling in ALS: 1 -Establish a culture on a synthetic wastewater, acclimating the biomass to an influent concentration of 1000 mg/L 2 -Feed the culture the condensate from the STAR reactor when the shipments are received from Purdue 3 -Add chabazite and clinoptilolite to the acclimated biomass to determine if this enhances the capability of the mixed culture to oxidize ammonium to nitrate 4 -Determine the ability to perform nitrification then denitrification in the same system, completing the nitrogen cycle

Waste Treatment Using Tilapia: 1 -Quantify nutrient concentrations of tilapia 2 -Quantify degradation of fiber in tilapia by fully characterizing gut microflora and enzymes secreted by fish . 3 -Initiate evaluations of alternative production systems relative to traditional approaches. 4 -Continue characterizing nutrient concentrations in waste products. 5 -Quantify nutrient concentrations in plant waste products that had been used for fungal (mushroom) growth and the acceptance of those food products by tilapia. 6 -Quantify mineral flows from fish to plants.

Biosolids Dewatering: 1 - Identify biomass specie(s) best suited for dewatering and capturing nutrients in STAR waste effluent. 2 - Develop a cropping system that utilizes the dewatered biosolids form the STAR reactor to grow crops for food, mushroom growth and fish feed. 3 - Investigate plant growth substrates for STAR effluent dewatering process.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products. Add to knowledge about low-flush toilet operations

For STAR:
1. Initial pH adjustment and loading study, Dawn Whitaker: The treatment of raw wastes by the STAR system is a novel application of autothermal thermophilic aerobic digestion (ATAD). In the first experimental phase, a reactor was designed scaled for 1-3 crew members to provide proof of concept. HRT was varied from 9-18 days, and solids concentrations of the influent varied from 4-10%. Total and volatile solids destruction was evaluated, and pH and ORP utilized as performance monitors to determine optimal solids loading.

In the fall of 2004, it was determined that the initial pH levels present in the first few hours of start-up were critical for the success of the start-up. Due to a change in feedstock formulation, the influent pH was lowered. This lower initial pH prevented the successful start-up of the reactor. Conditions remained hampered in the system, reflecting acid hydrolysis rather than the desired micro-aerobic digestion. Degradation rates were lower, achieving only ~50-55% total solids degradation, and pH levels remained around 4, which would negatively impact downstream uses of the effluent for projected plant growth. Early in 2005, an in-depth study of the effects of pH and solids loading began.

The reactor was maintained at 55-60C during the study. Aeration was provided using a stainless steel diffuser submerged in the reactor tank with airflow of 2.5 L/min. An air diaphragm pump provided circulation and mixing. The reactor was batch fed on a daily basis. HRT for the study was maintained close to 10 days. The one exception to this was a trial at 8% solids loading with an HRT of 18 days as well as 10 days to determine the effect of HRT on the solids loading deemed most likely to succeed. Successful start-up of the system was determined to be dependent on the utilization of pH control (using NaOH) for the first several hours of the run. As shown in the representative graph (2% solids loading), the system was able to become self-regulating after the first 8-12 hours, after which time no further addition of NaOH was needed. This result was found to be consistent at all solids loadings evaluated

The other crucial parameter evaluated was solids loading. Both aeration and mixing are dependent upon the influent solids levels. Due to pump constraints, it was hypothesized that the optimal solids influent level would be 10%. Optimization of the influent solids loading showed the highest degradation (75%) at an influent solids loading of 6%. However, there was only a minimal loss of efficiency at 8% loading (71-73%). Degradation was decreased to 60% total solids loss at the 10% influent loading.

Successful start-up will likely require the addition of base to maintain pH levels until the system becomes self-regulating. The optimal influent solids loading level was determined to be 8%, with a HRT of 9 days. In order to optimize the system, mass of both the reactor and contents must be reduced, and the addition of water should be minimized. By increasing the solids loading from 6 to 8%, 700 mL of water is conserved per crew member per day. Over the course of a long-term mission this reduction is substantial. Further water conservation could be achieved by increasing the solids loading to the 10% range; however, these preliminary studies at 10% solids loading showed relatively unreliable performance of the system as well as a significant reduction in degradation. Finalization of results at the 10% loading is ongoing at the time of this writing, and will be completed in early October.

2. Composting study, Angela Nolan: Based on the inedible plant biomass generated by the all-crop diet, 6.7 kg of inedible plant biomass would be generated per person per day (4). This is an enormous quantity of plant matter, and presents extreme difficulty with regards to pumping in the STAR system. It was therefore decided to add an additional chamber specifically for the treatment of inedible plant matter beyond that of the salad machine crops for this type of scenario. Day-to-day operation of the STAR reactor would include human fecal matter, toilet paper, food residuals, and the inedible biomass generated by the salad machine crops. The additional chamber would only be utilized for treatment of large quantities of crop biomass.

Initial research focused on finding an optimal method to treat the plants prior to composting. A preliminary study was to evaluate whether the size of soybean plants affected their ability to degrade during composting or not. It is a common notion that smaller particle size enhances degradation as opposed to larger sizes, but the question is “how small is small?” The time it takes to cut the plants increases as the particle size needed decreases. By testing the ability of different sized soybean plants to degrade, we hoped to use the results to develop a pre-processing method that would incorporate optimal degradation conditions for the plant and lower work requirements to help reduce ESM calculations.

Soybean stems and leaves were obtained and cut into 0.5, 1.0, and 1.5-2.0 centimeter squares using scissors. A mass of each size was placed in a mini-composter that consisted of an air-tight plastic container with a mesh screen in the middle. The soybeans were placed on top of the screen (to allow liquid to fall through), and three valves allowed for the addition of air, removal of air, and the removal of excess liquids. The composters were then placed in a temperature-controlled vessel containing approximately three inches of water and a submersible heater that maintained temperatures at 48?C. Each day, the gases in the composter headspace were evaluated for oxygen and CO2 using a Rosemount Analytical O2 and CO2 Analyzer. The carbon dioxide and oxygen peaks were used to evaluate activity within the composter based on the fact that oxygen is consumed and carbon dioxide is produced during composting.

This experiment was only conducted for one month, and little data was obtained during that time as the focus was on improving the physical set-up. Although there is not adequate data to show a correlation between particle size and degradation capacity, it was concluded that the experimental setup has the capacity to produce such data when the experiment is allowed to run throughout its duration.

3. Bench-scale filtrate recycling and enzyme activity study, Dawn Whitaker: An evaluation of the potential use of STAR effluent filtrate to enhance degradation and reduce the amount of clean water required by the system to adjust influent solids levels is currently being completed. It is hypothesized that degradation within the STAR system will be increased by recycling filtrate from the treated sludge effluent back into the influent stream. An additional benefit will be the reduction in the amount of clean water required for operation, which will benefit the ESM of the system. The premise of the study is to compare the effects of hot water, effluent inoculum and filtrate inoculum on the degradation of both STAR feedstock and newsprint. The lignin content in the feedstock will be varied by the use of a lower lignin content paper along with newsprint. Unadulterated STAR effluent will be evaluated in the system to determine any further degradation potential and CO2 generation.

STAR effluent is filtered sequentially through a 10- and 1-micron filter to separate solids. The analyses utilize a Columbus Instruments respirometer, which monitors and controls CO2 and O2 levels within the headspace of the 1-L

vessels. The system is currently equipped to run 3 samples in each run. Samples are well-mixed and maintained at 55?C. A factorial experimental matrix will be tested, evaluating each of the components separately (filtrate, solids, feedstock, effluent) and all combinations of the components. Additionally, enzyme activity will be investigated in the individual solid and liquid components of STAR effluent. A time-scale study will be completed on the evolution of enzyme activity over the daily cycle of the system.

4. Lignin Degradation Study, Dawn Whitaker: Degradation of lignocellulose, a major component in the predicted feedstock for STAR, is under evaluation. This evaluation will enable optimization of the system as well as providing necessary information for potential downstream technologies. Influent and effluent lignin content along with CO2 generation and O2 consumption can be used to estimate degradation kinetics in the STAR system. These estimates can be utilized along with feedstock characteristics to predict degradation and optimize the desired parameters for STAR.

Lignin and cellulose/hemicellulose degradation are key to reducing waste volume and achieving high degradation rates both in space mission scenarios as well as on earth. Microbial cellulose utilization is responsible for one of the largest material flows in the biosphere and is of interest in relation to analysis of carbon flux at both local and global scales (Lynd et al. 2002b). Lignocellulose constituents are major components in ALS solid wastes. Feces contain approximately 30% lignocellulose, while plant matter and food residuals can contain up to 70-80% (Komilis and Ham 2003). Huge quantities of plant growth are projected for habitat scenarios – 6.7 kg/cm-d of inedible biomass is estimated for full crop diets (Hanford 2004). Optimization of solid waste systems to achieve high degradation rates of these substances and recover water, carbon, and nutrients will be crucial.

ASTM 1721 Determination of Acid-Insoluble Residue in Biomass is the method used to assess lignin content. Samples from both the STAR pilot-scale reactor as well as the filtrate bench-scale studies (See #3 above) are being assessed for lignin degradation. Preliminary studies on lignin degradation in the pilot-scale STAR system have shown degradation rates of approximately 80%. Published values for biological degradation of lignin range from 0-70%.

For PAABLO/WAABLO: Research has just commenced. Therfore, no results to report at this time.

For Nitrogen Cycling: Completion of Zeolite Adsorption Studies. Ammonium removal by clinoptilolite has been found to be a cost effective method in wastewater treatment over the last three decades. Clinoptilolite, a naturally occurring zeolite, has been found to be selective for ammonium ions in the presence of other cation concentrations commonly found in wastewaters. Five zeolites including two types of clinoptilolite and two types of chabazites were purchased from GSA Resources, Tucson Arizona. The last of the zeolites was a clinoptilolite from Bear River Zeolite Company (BRZ) Thompson Falls, Montana. The purpose of the project was to find a suitable means of decreasing the concentration of ammonia-rich brine fluid, which includes some organics and hydrogen sulfide from condensate from the STAR Gas Scrubber.

The subject of study was the equilibration of ammonium ion on to three zeolites in the presence of sodium and potassium ion, when treated with simulated wastewater (1000 mg NH3 - N/L) in a batch reactor setup. The chabazite ZS500RW/H was found to be the most stable and had a maximum adsorption capacity at equilibrium (32.73 mg NH3-N/ gzeolite) as compared to the commonly used clinoptilolite. The chabazite at an initial concentration of 1000 mg NH3-N/L attained 90% removal in less than one hour with only one gram of the respective zeolite. However, a maximum adsorption capacity was higher for clinoptilolite ZS403TM (36.20 mg NH3-N/ gzeolite) in comparision to that of ZS500RW/H (34.40 mg NH3-N/ gzeolite). But the most important property of retention capacity at equilibrium differs largely in the case of ZS403TM (25.82 mg NH3-N/gzeolite) than in the chabazite. The adsorption capacity of clinoptilolite ZK406H was also found to be consistent with the general capacity of clinoptilolites in the literature of 18 mg NH4+/g of zeolite.

A column study was also performed with synthetic wastewater to determine the breakthrough curves for the three zeolites. The chabazite ZS500RW/H was found to outperform the other forms of zeolite which was consistent with the results of the batch reactors. This may be due to the presence of easily exchangeable sodium ions. Further, Ming et. al, 1995 and Lahav et. al, 2000 observed that the affinity of clinoptilolite and chabazite was greater for K+ than NH4+. This affinity for potassium cations over ammonium can explain the poor adsorption of ammonia for the other zeolites.

The optimum zeolite screened from the above experiments was tested over a range of concentrations of ammonia (Figure 3) to determine its maximum capacity for lower concentrations and behavior over various concentration ranges. At the 400 mg NH3-N/L concentration the retention ability of ZS500RW/H for ammonia was comparable with a maximum adsorption of 22.57 mg NH3-N/gzeolite to an equilibrium adsorption capacity of 21.81 mg NH3-N/gzeolite. It was observed that an increase in initial concentration 500 and 1000 mg NH3-N/L resulted in an effectual increase in the adsorption capacity, 31.93 mg NH3-N/gzeolite and 40.87 mg NH3-N/gzeolite for ZS500RW/H, respectively. However these values were misrepresentative since the equilibrium adsorption capacities were 66% and 63% respectively, of the original aqueous ammonia retention.

To increase the adsorption capacity by adjusting the various parameters responsible for adsorption, different pretreatment techniques were carried out for the best zeolite and at 400 mg/L concentration of the ammonia solution, as shown in Figure 4. It was found that the heat treatment for one hour at 600 ºC showed the best results. Based on the observations of Klieve et. al (9), the heat treatment for one hour was carried out. This heat treatment resulted in weight loss due to evaporation of water and some organics which opened channels for better ammonia adsorption. During the initial five hours, the untreated zeolite ZS500RW/H had the best adsorption capacity but maintained a very low retention value, while the one hour heat pretreated sample had comparative adsorption and retention capacities. Researchers have observed that Na-exchanged natural chabazite showed solid-state transformation to a sodalite structure at 600 ºC ,whereas K-exchanged natural chabazite did not transform. However, the XRD spectra taken after the heat treatment of the samples under study did not show any such structural changes as compared to untreated samples

A pretreatment process involving an acid treatment may result in dissolving alkali impurities which can block pores and channels within the zeolitic framework, hence promoting adsorption deep within the pores. However, zeolites with lower Si/Al ratio have been found to be unstable at lower pH, probably due to the loss of aluminum from the framework and thus, resulting in lower adsorption capacity. At an initial concentration of 400 mg NH3-N/L, the untreated zeolite performs better than the acid pretreated. Another pretreatment aimed at increasing the ammonia adsorption capacity of the zeolite was treatment with sodium hydroxide. Heating in the presence of salt for varying time periods did not improve the adsorption capacity. For instance, ZS500RW/H base treated had a maximum adsorption capacity (27 mg NH3-N/gzeolite) that surpassed that of the untreated zeolite (15 mg NH3-N/gzeolite) and one hour heat treated also. But the retention capacity of one hour heat treated chabazite was the best among all the other pretreated samples.

Task Progress:

When the initial NH3-N concentrations applied to the batch reactors was very low, ammonium uptake by the clinoptilolite zeolite was comparable to that of the chabazite zeolite. However for very high loadings of NH3-N concentrations, chabazite with sodium ion as exchangeable ion was found to be the best, in terms of the adsorption capacity. The adsorption capacity of chabazite was enhanced by pre-heating for one hour.

Initial Nitrification/Denitrification Studies in Sequencing Batch Reactors. The objective of this phase of the study was to evaluate the ability of nitrifying and denitrifying bacteria to process ammonia to nitrogen gas. In order to investigate the bioavailability of precipitated phosphorus, SBR bioavailability experiments were conducted. During this project, we were confronted by various operational issues. This report presents those issues and also proposes some solutions.

Sequencing Batch Reactor System. Two bench scale sequencing batch reactors (SBR) running in parallel were used to perform nitrification/denitrification process by providing aerobic followed by anoxic conditions. To ensure cycling between aerobic and anoxic conditions the two reactors, an airtight fitting was delivered air into the sealed reactors. Two motor driven mixers provided mixing during the nitrification/denitrification process. Two pH and DO probes were used for on-line pH and DO measurements using a Hydra Data logger Unit. Three double-headed pumps were used to fill and decant the SBRs and to add sodium bicarbonate as a base to the reactors in order to adjust the pH to 7. A Chrontrol timer was used to control the initial 12-hour SBR cycle which comprised of a total fill time of 15 minutes, a reaction period of 7 hours for nitrification and 4 hours for denitrification, a settling period of 30 minutes and decant time of 15 minutes. Each reactor was seeded with 20 L of activated sludge from Blue Plains WWTP tertiary treatment process. The collected activated sludge placed in the reactors was acclimated under experimental conditions to room temperature at 22-25 ºC.

Synthetic wastewater consisting of condensate water scrubbed from the gas of the STAR system was going to be used with an ammonia nitrogen concentration and a phosphate phosphorous concentration. Initially synthetic wastewater was utilized in place of feed from the STAR system feed and to acclimate nitrifiers and denitrifiers in the two reactors. When nitrification and denitrification take place, with the large amount of methanol, measured as soluble COD, that is necessary to complete denitrification. The precipitous drop in COD at the end of nitrification is believed to be the result of aerobic heterotrophic use of the organic carbon with the dissolved oxygen present at the end of nitrification. Results confirm that a drop in dissolved oxygen with the shift from aerobic to anoxic conditions occurred during the seventh hour of the reaction cycle.

Results confirm that ammonia nitrification is complete within three hours and that there is no build up of nitrite during the reaction. During denitrification nitrite does accumulate during the reaction and is completely denitrified by the end of the reaction period. During this particular profile oxygen entered the system to interfere with the completion of the reaction, and cause the nitrification of nitrite back to nitrate. This preliminary experiment has confirmed that nitrification followed by dentrification is possible in Sequencing Batch reactors. A delicate balance between aerobic and anoxic conditions must be maintained in this type of single consortium system. We are currently increasing our initial concentration following the protocol established in the experimental design.

For Tilapia Use in Waste Treatment:

Approximately 2 weeks prior to starting the third year of the NSCORT project (November 13, 2004), the Aquaculture Research Laboratory was completely destroyed by fire. Temporary facilities have been occupied, fish have been reacquired and experimental systems have been reestablished, but we were obviously delayed in starting research this year.

Nutrient concentrations in tilapia. We completed our initial evaluations of nutrient concentrations in tilapia. We conducted a near exhaustive nutrient profile for both whole fish and fillets. Fatty acid concentrations , amino acid concentrations and a comparison of crew nutrient requirements to the nutrient concentrations found in tilapia were studied. Amino and fatty acid requirements for crews are not presented as those values have apparently not been developed. Based on these data, tilapia should be considered a high-quality food for crews.

Fiber Degradation. We initiated a study in which we are quantifying gut microflora and gastrointestinal tract enzymes present in tilapia fed fiber concentrations ranging from 0 to 20% of the diet. Two new collaborations were established with this project. Dr. Ching-Ching Wu, Purdue, Animal Disease Diagnostic Laboratory is a bacteriologist with a fully equipped laboratory for quantifying both aerobic and anaerobic bacterial populations, and Dr. Jiri Adamec is a lead scientist in the newly established Bindley Biosciences Complex. The Bindley Complex has individual Genomics and Proteomics Centers. Our work will be conducted in the Proteomics Center. We will collect samples and characterize bacterial flora as a function of fiber ingestion in tilapia. Additionally, we will isolate enzymes in the gastrointestinal tract as a function of fiber ingestion. Samples will be collected from three segments of the intestine and proteins separated with one- and two-dimensional gel electrophoresis. Treatment differences will be determined with a gel imaging system and those spots will be harvested for protein characterization and quantification using LC-MS. This will be an important step in understanding fiber degradation in tilapia.

Alternative Production Systems. The integrated aquaponics systems, as developed and used on Earth, are large and cumbersome. We are exploring the potential of altering the flow of nutrients within a fish production system from the integrated aquaponics system to a stand-alone fish production system utilizing heterotrophic bacteria in combination with nitrifying bacteria. The modified system appears more efficient and our initial trial is underway. We are comparing an integrated system with a modified heterotrophic fish production system. Growth of fish as a function of nutrient inputs and resulting water quality (dissolved oxygen, pH, ammonia-N, nitrite-N, nitrate-N, chemical oxygen demand, and alkalinity) are the primary responses being quantified in this initial evaluation.

Characterization of nutrient concentrations. We recently acquired a gas chromatograph from our insurance settlement that we will use to quantify fatty and amino acids in waste products as they become available from collaborators evaluating crop species. Based on our initial evaluations of waste products, we identified those nutrient categories as limiting for tilapia growth. Fatty acids will be separated and quantified using standard methods published by the American Oil Chemists' Society employing methyl esterification of individual fatty acids. Unknowns will be compared to standards acquired from commercial sources. We will use a new method for separating and quantifying amino acids with GC and flame ionization detection. The EZ-FAST method was recently released by Phenomenex and we will be working with them to validate their new method with our specific tissues. The basics are similar to all other amino acid analysis methods in that proteins are hydrolyzed, derivatized then separated. The obvious difference is that amino acids are separated using gas as the mobile phase instead of organic solvents. Given our earlier results indicating tilapia will

consume a wide range of waste products, we feel nutrient characterization is sufficient for the next year. A final feeding study will probably be warranted once collaborators finalize decision regarding candidate species of plant. Our initial fatty and amino acid analyses are being conducted with soybean residue that has been used as a substrate for fungal growth. Working with Drs. Mitchell and Beyl, we acquired soybean residue, sent those to Dr. Beyl where she inoculated the residue with fungi. We acquired treated and untreated samples and are conducting the analyses now.

Fungi treated crops as food for tilapia. Due to limited biomass, we were only able to quantify the nutrient concentrations of soybean residue treated with fungi. We acquired inoculum from Dr. Beyl and treated a separate sample of cow pea waste, collected that product and are feeding treated and untreated cowpea waste to juvenile tilapia. Full nutrient characterization and quantification is underway. Biological responses from fish will also be quantified. This project will help direct the flow of fibrous plant residues within the ALS to maximize degradation and efficient production of food.

Quantify mineral flows. The mass balance of minerals from feed or waste products, through fish then into solution for plant use is one of the keys areas in aquaponics. We established seven aquaponics systems with fish and lettuce, then modified diets to test the limiting nutrients for plants from the fish feeds. Boron, iron and manganese appear to be the limiting nutrients for plants. We established a positive control, a negative control, and test diets with each mineral alone or in combination. This study was lost in the fire that destroyed the Aquaculture Research Laboratory and, using that well water source, all minerals were deficient for plants. In our temporary laboratory, manganese concentrations are sufficient for plant growth, so we are limited in the tests we can run. We are feeding fish in that study and will replicate the study once we move into our reconstructed building (anticipated completion date January 2006). Our goal is to quantify mass balance in a static system so our evaluation of systems can be more complete than previous efforts.

For Biosolids Dewatering:

Experiment 1: Candidate species for dewatering of biosolids include grasses, legumes, and food crops from the NASA ALS program. Plants of each species were established in 1 L pots containing coarse silica sand and established using a complete nutrient solution during two months of establishment in the greenhouse. Legumes were inoculated with the appropriate strain of Rhizobium to facilitate symbiotic fixation of atmospheric dinitrogen gas into plant-available forms. Pots were placed in a greenhouse set to control temperature (25ºC±5ºC) and photoperiod was extended to 15 h with artificial lighting. After establishment, pots were grouped into three replicates each containing four pots of each species. The following four STAR effluent treatments were randomly assigned to pots of each species within each replicate: control (no effluent); 1/2 acre-inch of effluent (86 mL per 1 L pot); 1 acre-inch of effluent (172 mL per 1 L pot); and 2 acre-inches of effluent (344 mL per 1 L pot). Cups were placed under pots to catch liquid that drained from the bottom of the pots following effluent application. Pots remained in the greenhouse and were watered with reverse-osmosis water as needed. Photos of plants were taken one week after effluent application. Plants were destructively sampled two weeks after effluent application to obtain dry weights of roots and shoots, and tissues for mineral analysis. Poor establishment prevented inclusion of three replicates of Eastern Gamagrass. Both wheat cultivars developed very rapidly, and were setting seed after two months so they were not included in this study, but were evaluated in a subsequent study.

During the week following effluent application there was no visible injury to any species caused by the ½ acre-inch effluent application. Species exhibiting the least injury included kura clover and switchgrass, while tall fescue and reed canarygrass exhibited mild and moderate injury, respectively, to at the 2 acre-inch effluent rate. Leaf tips of rice were necrotic on plants provided 1 and 2 acre-inch effluent rates, but otherwise these plants were relatively uninjured. Several species receiving 2 acre-inches of effluent were severely injured including big trefoil, Ladino clover, and pepper. Tomato appeared to be most sensitive to effluent application with severe injury exhibited by plants provided 1 and 2 acre-inch effluent treatments.

Experiment 2 Candidate species for dewatering of STAR effluent in Experiment 2 include grasses, legumes, and food crops. Three plants of each of the eleven candidate species were established in 1 L pots containing a coarse silica sand and provided nutrient solution for four weeks. Pots were placed in a greenhouse set to control temperature (25ºC±5ºC) and photoperiod of 15 h with artificial lighting. Plants growing in 1 L pots were then placed in a second 1 L pot for submersion into one of three treatments (STAR effluent, nutrient solution, and deionized water). All three treatments were delivered via a tube inserted into the bottom 1 L pot. Treatments were delivered daily to each plant specie, recording volume for water use and efficiency data. After twenty five days, plants were destructively harvested and divided into herbage, crown and root components and analyzed for B, Ca, Cu, Fe, K, Mg, Mn, N, NO3, Na, P, S, and Zn composition.

Water transpiration increased as plant mass increased for water and Hoagland's treatments, but not with the STAR treatment. All species grown in STAR effluent exhibited reduced growth and enhanced senescence. Hoagland's nutrient solution generally increased tissue concentrations for most elements tested. Tissue sodium concentrations of plants grown with STAR effluent were significantly greater than that observed with the other treatments.

Experiment 3 The following candidate species were included for dewatering STAR effluent in Experiment 3. These plants were selected based on their potential for growth in the anoxic effluent environment. Cattail Rush Kopu II White Clover Reed Canarygrass Super Dwarf Rice

Three plants of each of the five candidate species were established in 1 L pots containing an illite-montmorillinite-silica substrate (Turface) and provided nutrient solution for four weeks. Pots were placed in a greenhouse set to control temperature (25ºC±5ºC) and photoperiod of 15 h with artificial lighting. Plants growing in 1 L pots were then placed in a second 1 L pot for submersion into one of two treatments (STAR effluent and nutrient solution). Treatments were administered in the same fashion as in experiment 2. After one week, the STAR treatment was substituted with deionized water due to limited STAR effluent availability. Plants were destructively harvest after four weeks and divided into herbage and root components for dry weight analysis.

Cattail was the only specie that outperformed the nutrient solution treatment in terms of total dry weight production. Cattail also was the only specie observed to have roots actively growing in STAR effluent. Total dry matter production of all other species was less than that of the nutrient solution treatment.

Planned Research for Biosolids Dewatering in 2005: Based on poor plant performance when placed in the STAR effluent and EAC feedback, we had planned to work with Dr. Alleman's group to dewater the STAR effluent via freeze-drying or direct physical solid-liquid separation, mixing the solids with an inert and recoverable/recyclable silica glass root growth media, and use food crops to recover and recycle nutrients from the STAR residuals.

	This proposed change in direction was sent to the project director in February 2005. In February 2005, the project director unilaterally eliminated this project from the program because he wanted “to de-emphasize the role of plants in waste processing in space habitats and to go in a different direction.” This was the only meeting we ever had with the project director during this project. Funding was provided only to partially cover trainee costs and close out costs during 2005, so no additional experiments were conducted in 2005. One of the greatest challenges we faced throughout this project was the lack of available STAR effluent. Dr. Alleman requested funding to build a second reactor in 2004, but our understanding was that his request was denied by the project director.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Howard SM, Volenec JJ, Joern BC, Alleman JE, Whitaker DR. "Biosolids Dewatering of Solid-Phase Thermophilic Aerobic Reactor (STAR) Effluent Using Various Plant Species." American Society of Agronomy Annual Meeting, Seattle, WA, Nov 2004. Agronomy Abstracts. November 2004. , Nov-2004
Articles in Peer-reviewed Journals	Gonzales JM, Hutson AH, Rosinski ME, Brown PB, Wu YV, Powless TF. "Evaluation of fish meal-free diets for first feeding Nile tilapia, Oreochromis niloticus." Journal of Applied Aquaculture. 2007 Aug;19(3):89-99. http://dx.doi.org/10.1300/J028v19n03_06 , Aug-2007
Articles in Peer-reviewed Journals	Gonzales JM Jr, Brown PB. "Nile tilapia Oreochromis niloticus as a food source in advanced life support systems: Initial considerations." Advances in Space Research. 2006;38(6):1132-7. http://dx.doi.org/10.1016/j.asr.2005.11.002 , Sep-2005
Articles in Peer-reviewed Journals	Glass CC, Chee Wah R. "Enhancement of Ammonium Adsorption Capacity with Pretreated Zeolites." Habitation, July 2005. , Jul-2005
Papers from Meeting Proceedings	Glass CC, Chee Wah R. "Nitrogen Recovery during Solid Waste Treatment in Advanced Life Support." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems, Paper #04-01-2514. July 2004. , Jul-2004
Papers from Meeting Proceedings	Whitaker DR, Staton KL, Alleman JE, Lane JW. "Loading Balance and Influent pH in a Solids Thermophilic Aerobic Reactor." International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, July 2005. International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, Paper # 2005-01-2982. July 2005. , Jul-2005
Papers from Meeting Proceedings	Kuo Y, Whitaker D, Chiu G, Alleman J. "System Level Design and Initial Equivalent System Mass Analysis of a Solid-Phase Thermophilic Aerobic Rector for Advanced Life Support Systems." International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, July 2005. Conference Proceeding: International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, Paper # 2005-01-2983. July 2005. , Jul-2005
Presentation	Whitaker, D.R., Alleman, J.E. "Thermophilic Aerobic Solid Waste Processing for Long-term Crewed Missions." Earth and Space 2006, 10th ASCE Conference on Engineering, Construction, and Operations in Challenging Environments, Houston, TX Jan-2006
Presentation	Whitaker, D.R. and Alleman, J.E. "Lignin Degradation in Solid Waste Treatment Systems." Habitation Conference, American Institute of Aeronautics and Astronautics, Orlando, FL. Feb-2006
Presentation	Whitaker, D.R "Solid Waste and Biosolids in Space" Chemistry, Bioavailability, and Toxicity of Constituents in Residuals and Residual-Treated Soils Group Meeting,Las Vegas, NV. Jan-2005
Presentation	Glass, C. C. and R. Chee Wah "Nitrogen Recovery during Solid Waste Treatment in Advanced Life Support." Presented at the Habitation Conference, Orlando, FL. Feb-2004
Presentation	Sinclair, H "Ammonium adsorption by ion exchange using pretreated zeolites," Presented at the NSCORT Summer Symposium, Purdue University, West Lafayette, IN. Jul-2004
Presentation	Gonzales , J.M., Y.V. Wu, T. Powless and P.B. Brown. "Evaluation of fish meal-free practical diets for first feeding Nile tilapia Oreochromis niloticus. " Aquaculture 2004, Honolulu, HI. Feb-2004
Presentation	Alleman, James E., P.B. Brown, C.C. Glass, B.C. Joern and J.C. Volenec "Coordinated Biochemical Stabilization and Resource Recovery with Human-, Plant-, and Food-Derived ALS Solid Wastes" Habitation 2004, Orlando, FL. Feb-2004
Presentation	Gonzales, J.M., Jr. and P.B. Brown. "Quantitative and qualitative characterizations of the nutritional impact Nile tilapia Oreochromis niloticus may offer in an advanced life support system. " Aquaculture America 2005, New Orleans, LA. Jun-2005